

Multiple Choice (10 marks)

1. How does a buffer overflow on the stack facilitate running attacker-injected code?
 - (a) By overwriting the return address to point to the location of that code
 - (b) By writing directly to the instruction pointer register the address of the code
 - (c) By writing directly to %eax the address of the code
 - (d) By changing the name of the running executable, stored on the stack
2. Which of the following is true of mutation-based fuzzing?
 - (a) It generates each different input by modifying a prior input
 - (b) It works by making small mutations to the target program to induce faults
 - (c) Each input is mutation that follows a given grammar
 - (d) It only makes sense for file-based fuzzing, not network-based fuzzing
3. Can we build a secure PRP from a secure PRG?
 - (a) No, it cannot be done
 - (b) Yes, just plug the GGM PRF into the Luby-Rackoff theorem
 - (c) It depends on the underlying PRG
 - (d) Option text
4. A special tool is necessary for entering the network which is _____ that helps the anonymous internet users to access into the Tor's network and use various Tor services.
 - (a) Opera browser
 - (b) Firefox
 - (c) Chrome
 - (d) Tor browser
5. Can a stream cipher have perfect secrecy?
 - (a) Yes, if the PRG is really "secure"
 - (b) No, there are no ciphers with perfect secrecy
 - (c) Yes, every cipher has perfect secrecy
 - (d) No, since the key is shorter than the message

True / False (10 marks)

Answer True or False

6. True or False: Weak or non-existent authentication mechanisms are not considered presentation layer issues in cybersecurity.
7. True or False: Backdoor Trojans primarily focus on stealing personal data without incurring any financial costs.
8. True or False: EXE restarts the STP solver and retries the constraint query indefinitely until it finds a satisfiable result.
9. True or False: In message confidentiality, the transmitted message must be clear to both the sender and the translator.
10. True or False: The one-time pad key can be computed by using the formula $k = m \text{ xor } m$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the demlo number for the given number.
12. Write a function to find all adverbs and their positions in a given sentence.

End of Paper